

Churn Analysis & Customer Intelligence

End-to-end OTT retention analytics using SQL + Python

- Multi-table customer, subscription & support intelligence
- 20+ business KPIs • Risk segmentation • Revenue impact
- Portfolio-ready executive analytics

1. Business Challenge

Retention is the growth engine in a competitive OTT market.

- Identify who is most likely to churn.
- Understand what customers did before leaving.
- Find the customer “danger zone” where intervention matters most.
- Quantify revenue and CLTV exposure—not just churn percentage.
- Turn analysis into targeted retention actions.

28.6%

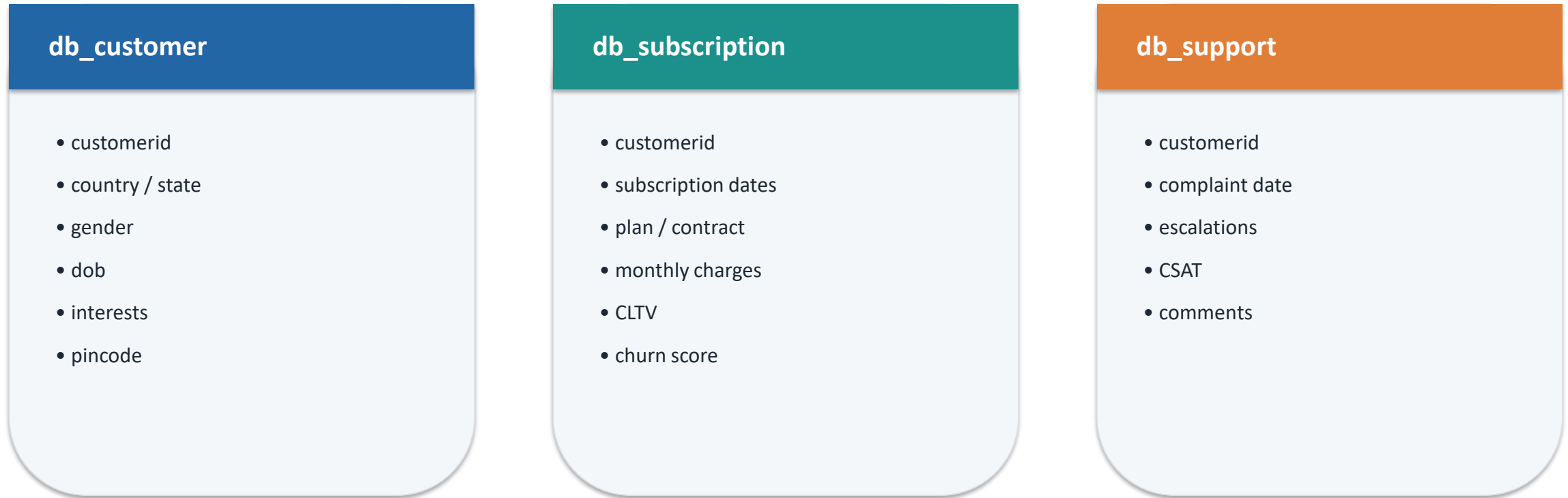
Overall churn rate

71.4%

Retention rate

2. Data Architecture

Relational data model connects customer identity, commercial behavior and service experience.



JOIN ON customerid

3. Analytical Workflow

A reproducible SQL → Python → insight pipeline.



4. Executive KPI Scorecard

Metrics designed to connect customer behavior with financial impact.

28.6%

Churn Rate

71.4%

Retention Rate

₹18.8

ARPU

1,451

Avg. Tenure (days)

₹395

Total Revenue

₹73.94

Revenue at Risk / MRR

₹2,047

CLTV Lost

18%

Revenue Loss

5. Key Finding — Contract Structure

Monthly subscribers represent the clearest retention opportunity.

55.6%

Monthly-contract churn

8.3%

Annual-contract churn

6.7×

Relative churn rate

- The monthly segment is the dominant churn-risk cohort.
- Annual contracts show materially stronger retention.
- Prioritize contract migration and early-tenure engagement.
- Measure success through retained MRR and CLTV—not only customer count.

6. Where Churn Concentrates

Temporal and geographic signals identify where to investigate first.

Sep 2024

Peak churn month

Karnataka

Most affected state

Basic

Highest churn volume

- Investigate September events: pricing, product incidents, content changes or campaign effects.
- Deep-dive Karnataka for support escalations, complaints, CSAT and competitor movement.
- Basic-plan churn is high in volume, but revenue impact must be assessed before over-investing.
- Use state × plan × contract cohorts to isolate the true driver.

7. Support Intelligence

Service friction becomes a retention signal when escalations and churn move together.

- Join complaint, escalation and CSAT records with subscription outcomes.
- Compare churn rate for customers with ≥ 1 escalation vs customers with 0 escalations.
- Use CSAT and complaint frequency to prioritize intervention intensity.
- Decompose cancellation reasons into competitor switching, pricing sensitivity and content dissatisfaction.

Escalations

High-priority service signal

CSAT

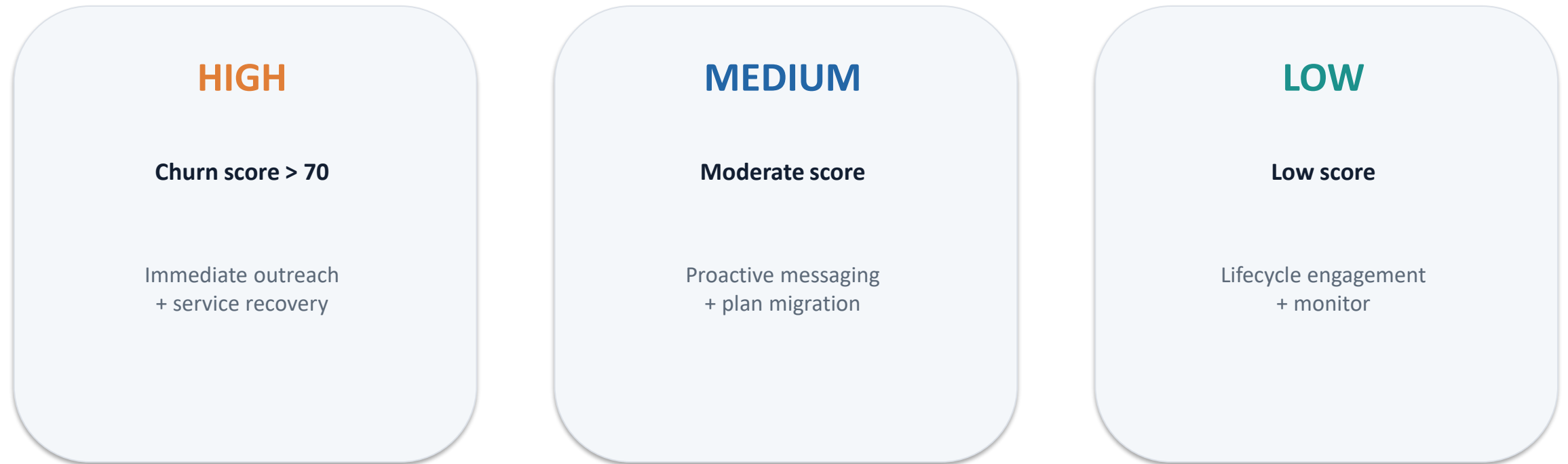
Customer experience signal

Reason

Root-cause signal

8. Risk Scoring & Segmentation

Composite risk turns descriptive analytics into an actionable retention queue.



- Prioritize high-risk customers by CLTV and support friction.
- Target Premium annual-plan retention where lifetime value is highest.

9. Recommended Retention Strategy

Move from “who churned?” to “who should we save first?”

- | | | |
|----|----------------------------------|--|
| 01 | Investigate Karnataka | Validate price, service, content and competitor signals. |
| 02 | Protect September cohorts | Audit incidents/campaigns and contact affected customers. |
| 03 | Migrate monthly users | Offer annual-plan incentives to high-risk, high-LTV customers. |
| 04 | Fix escalated cases | Create service-recovery workflows tied to churn risk. |
| 05 | Prioritize by CLTV | Allocate retention budget where saved revenue is highest. |

Portfolio Project — Executive Summary

- Engineered an end-to-end OTT churn analytics pipeline using SQL + Python across three relational tables.
- Built 20+ KPIs covering churn, retention, ARPU, tenure, revenue-at-risk, CLTV and support intelligence.
- Identified a 6.7× churn gap between monthly and annual contracts.
- Quantified ₹73.94/mo in MRR leakage and ₹2,047 in CLTV erosion for six high-risk customers.
- Delivered a contract-migration and risk-prioritization strategy to focus retention resources on highest-value cohorts.

SQL + Python

Core analytics stack

20+ KPIs

Executive scorecard

Actionable

Retention strategy

10. Resume-Ready Impact

Concise story for a Data Analyst portfolio or interview.

Engineered an end-to-end OTT churn analytics pipeline using PostgreSQL and Python, integrating customer, subscription and support data to build 20+ KPIs. Identified a 28.6% overall churn rate and a 6.7× churn disparity between monthly and annual contracts, quantified ₹73.94/mo MRR leakage and ₹2,047 CLTV erosion among six high-risk customers, and translated risk segmentation into a targeted contract-migration and customer-retention strategy.

Tech: PostgreSQL • SQL • Python • Pandas • NumPy • Matplotlib • Seaborn